



Computer Networks

Introduction to Computer Networks

Why Computer Networks?

- Computer technology developed → **high-speed calculations** became possible.
- As **hardware cost decreased**, computers were installed at different locations worldwide.
- This created a need to **transfer data between computers**.
- Earlier methods such as **tape drives, floppy disks, etc.** were used for data transfer.
- However, these methods were **slow and inefficient**.

Data Transfer Between Sender and Receiver

For communication between **Sender (S)** and **Receiver (R)**, several problems need to be solved:

1. **Packetization of data**
2. **Common format** for data
3. Conversion of **binary data into signals**
4. Determining **who can access the communication channel**

Therefore, we need a **Network Model (NM)** that handles different issues involved in data communication.

Network Models

Two important network models are:

1. **OSI Model**
2. **TCP/IP Model**

OSI Model

OSI — Open Systems Interconnection

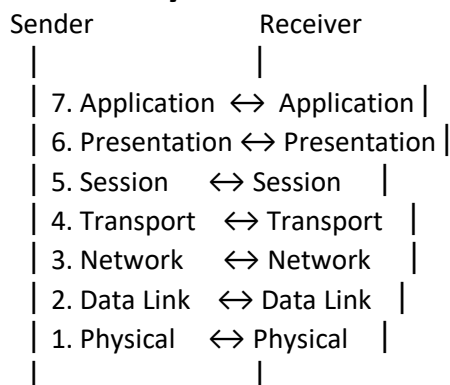
- OSI stands for **Open Systems Interconnection**.
- It was standardized by **ISO (International Organization for Standardization)**.
- The OSI model contains **7 layers**.

Meaning of "Open System"

An open system allows **two different systems to communicate**, regardless of their underlying:

- Hardware
- Software
- Architecture

Seven Layers of OSI Model



Layered Communication

Within a machine:

Each layer **uses the services of the layer above it** and **provides services to the layer below it**.

Between two machines:

Layer **n** of one machine logically communicates with layer **n** of the other machine.

These communicating entities at the same layer are called **peer processes**.

Peer

Peer means entities operating at the **same layer and following the same protocol/standard**.

Communication is generally **bidirectional**, so a sender can also become a receiver.

Encapsulation and Decapsulation

At Sender

As data moves **downward** through the OSI layers, each layer adds its control information.

Application

↓

M

Presentation

↓

H6 | M

Session

↓

H5 | H6 | M

Transport

↓

H4 | H5 | H6 | M

Network

↓

H3 | H4 | H5 | H6 | M

Data Link

↓

H2 | H3 | H4 | H5 | H6 | M | T2

Physical

↓

Signals

Where:

- **H** = Header
- **T** = Trailer
- **M** = Message/Data

At Receiver

The process is reversed.

Headers/trailers are removed as data moves **upward** through the layers.

This process is called **Decapsulation**.

Important GATE Point

Layer	Added Information
Application	No header/trailer
Presentation	Header
Session	Header
Transport	Header
Network	Header

Data Link Header + Trailer

Physical No header/trailer

Note: In practical protocol descriptions, the exact headers/trailers depend on the protocol. The above is the simplified OSI-layer representation.

Three Subgroups of OSI Layers

The 7 layers can be divided into three groups:

1. Network Support Layers

Layers 1, 2, 3

Layer 3 → Network

Layer 2 → Data Link

Layer 1 → Physical

These layers mainly deal with the **physical/network aspects of data transmission**.

2. User Support Layers

Layers 5, 6, 7

Layer 7 → Application

Layer 6 → Presentation

Layer 5 → Session

These layers mainly support **user-level communication and interoperability** between different systems.

3. Transport Layer

Layer 4 → Transport

- Provides **end-to-end delivery**.
- Responsible for delivering the complete message between end systems.

Layers 1, 2, 3 → Network Support

Layer 4 → End-to-End Delivery

Layers 5, 6, 7 → User Support

Functions of Layers

① Physical Layer

The **Physical Layer** is responsible for transmitting raw bits over the physical medium.

Functions

1. Physical Characteristics of Transmission Medium

Defines characteristics of the transmission medium:

- **Wired**
- **Wireless**

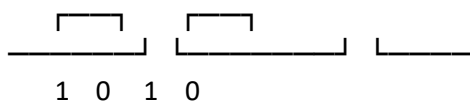
It deals with converting **binary data into signals** suitable for transmission.

Binary Data → Signals → Transmission Medium

2. Representation of Bits

Defines how **0 and 1** are represented as physical signals.

Example:



3. Data Rate / Transmission Rate

Transmission Rate (T.R.) = Number of bits transmitted per second.

$$T.R. = \frac{\text{Number of bits transmitted}}{\text{Time}}$$

Example:

If:

$$T.R. = 1000 \text{ bits/sec}$$

Then 1000 bits are transmitted in 1 second.

Therefore, time required for 1 bit:

$$\text{Bit Duration} = \frac{1}{T.R.}$$

$$= \frac{1}{1000} = 10^{-3} s$$

$$= 1 \text{ ms}$$

Important Formula

$$\text{Bit Duration} = \frac{1}{\text{Transmission Rate}}$$

and

$$\text{Transmission Rate} = \frac{1}{\text{Bit Duration}}$$

Transmission Rate vs Bit Duration

Transmission Rate Bit Duration

$$10^3 \text{ bps} = \mathbf{1 \text{ Kbps}} \quad 10^{-3} \text{ s} = \mathbf{1 \text{ ms}}$$

$$10^6 \text{ bps} = \mathbf{1 \text{ Mbps}} \quad 10^{-6} \text{ s} = \mathbf{1 \mu s}$$

$$10^9 \text{ bps} = \mathbf{1 \text{ Gbps}} \quad 10^{-9} \text{ s} = \mathbf{1 \text{ ns}}$$

$$10^{12} \text{ bps} = \mathbf{1 \text{ Tbps}} \quad 10^{-12} \text{ s} = \mathbf{1 \text{ ps}}$$

GATE Shortcut

Higher data rate → Lower bit duration

4. Synchronization of Bits

Sender and Receiver must be synchronized at the **bit level**.

The receiver must correctly identify:

- Beginning of a bit
- End of a bit
- Individual bit boundaries



5. Line Configuration

Defines how devices are connected to the communication link.

Point-to-Point

One dedicated link between two devices.

A ————— B

Multipoint

Multiple devices share a single link.

A ———— T ———— B

6. Physical Topology

Topology describes how devices and links are physically arranged in a network.

Examples:

- Bus
- Star
- Ring
- Mesh
- Tree

7. Transmission Modes

Defines the direction of data transmission.

Simplex

Communication occurs in **only one direction**.

S —————→ R

Example: Keyboard → Computer

Half Duplex

Communication occurs in **both directions, but one at a time**.

A —————→ B

A ←———— B

Example: Walkie-talkie

Full Duplex

Communication occurs in **both directions simultaneously**.

A —————→ B

A ←———— B

Example: Telephone call

★ GATE Question: Mesh Topology

Question

For a **full mesh topology** with n stations, how many point-to-point links are required?

Every station must be directly connected to every other station.

Number of links:

$$\boxed{{}^nC_2}$$

$$= \boxed{\frac{n(n-1)}{2}}$$

Example

For $n = 5$:

$$\frac{5(5-1)}{2} = \frac{20}{2} = \boxed{10 \text{ links}}$$

★ Remember

$$\boxed{\text{Full Mesh Links} = \frac{n(n-1)}{2}}$$

Because every pair of stations requires **one unique link**.

~Anshu